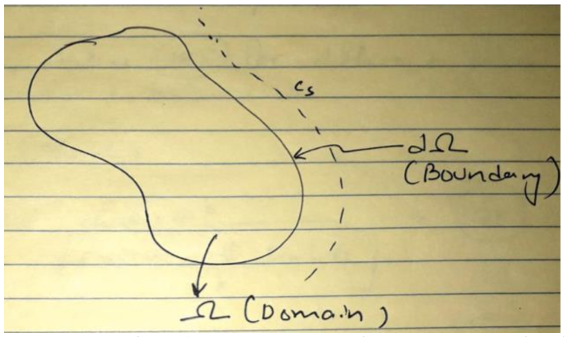
***Effect of Shape of Catalyst pellet on the Effectiveness Factor**

*The effectiveness factor for the first order reaction is readily obtained by solving the diffusion equation analytically.

*Consider a catalyst pellet of arbitrary shape with diffusion layer



Diffusion eqn in the domain is similar to 166, which can be written as

$$\nabla^2 u = \Phi^2 u \quad (175)$$

B.Cs

$$\underline{n}.\underline{\nabla} u = 0 \; at \; \rho = 0$$

$$\underline{n}.\underline{\nabla} u = v(1-u_s)\; \rho = 1 \quad (176)$$

Where

$$\nabla^2 = \frac{1}{\rho^a}\left(\frac{d}{d\rho}\rho^q \frac{d}{d\rho}\right) \quad (177)$$

Where ρ=0 (plate) ; ρ=1 (cylinder); ρ=2(sphere)

Solution to the (175) for different cases is summarized below

1. q=0 (plate)

$$u(\rho) = \frac{cosh\Phi\rho}{\cosh\Phi + \frac{\Phi}{v} sinh\Phi} \quad (178)$$

$$N(\Phi) = \frac{tanh\Phi}{\Phi + \frac{\Phi^2}{v} tanh\Phi} \quad (179)$$

$$v \to \infty, N(\Phi) = \frac{tanh\Phi}{\Phi} \quad (180)$$

2. q=1 (cylinder)

$$u(\rho) = \frac{I_0(\Phi\rho)}{I_0(\Phi) + \frac{\Phi^2}{v} I_1(\Phi)} \quad (181)$$

$$N(\Phi) = \frac{2I_1(\Phi)/I_o(\Phi)}{\Phi + \frac{\Phi^2}{v} I_1(\Phi)/I_o(\Phi)} \quad (182)$$

3. q=2 (sphere)

$$u(\rho) = \frac{sinh\Phi\rho}{\rho\left[\sinh\Phi + \frac{1}{v}(cosh\Phi - sinh\Phi - 1)\right]} \qquad (184)$$

$$N(\Phi) = \frac{3}{\Phi^2} \frac{\Phi\coth\Phi - 1}{\left[1 + \frac{1}{v}(\Phi coth\Phi - 1)\right]} \qquad (185)$$

$$v \to \infty, N(\Phi) = \frac{3\Phi coth\Phi - 1}{\Phi^2} \quad (186)$$

For each of the foregoing η for

$$\Phi \to \infty, N \to \frac{1}{\Phi}\ (plate);\ \frac{3}{\Phi}\ (sphere);\ \frac{2}{\Phi}\ (cylinder)$$

Again, assume that we have s species undergoing in chemical reaction

$$\sum_{j=1}^{s} \alpha_{ij} A_j = 0, i = 1,2 \dots n$$

Recall mole balance for the case of catalyst surface dispersed throughout the fluid

$$\frac{DC_j}{Dt} + \underline{\nabla}.\underline{J}_j = \sum_{i=1}^{m} \alpha_{ij}\hat{r}_i a_{vi} - \frac{C_j D ln\hat{V}}{Dt}$$

For the case of packed bed or fluidized bed

$$\epsilon \frac{\partial C_j}{\partial t} + \underline{\nabla}.\underline{N}_{j,e} = \sum_{i=1}^{m} \rho_b S_g \alpha_{ij}\hat{r}_i \quad (187)$$

The total flux:

$$\underline{N}_{j,e} = \underline{J}_{j,e} + C_j v_e$$

Neglect convective flux

$$\underline{N}_{j,e} = \underline{J}_{j,e}$$

From (187)

$$\epsilon \frac{\partial C_j}{\partial t} + \underline{\nabla}.\underline{J}_{j,p} = \sum_{i=1}^{m} \rho_b S_g \alpha_{ij}\hat{r}_i \quad (188)$$

From Fick's law

$$\underline{J}_{j,e} = \sum_{k=1}^{s} D_{jk,e} \underline{\nabla} C_k \quad (189)$$

Substitute (189) in (188)

$$\epsilon \frac{\partial C_j}{\partial t} = \sum_{k=1}^{s} \underline{\nabla}.\left(D_{jk,e}\underline{\nabla}C_k\right) + \sum_{i=1}^{m} \rho_b S_g \alpha_{ij} \hat{r}_i \quad (190)$$

***Energy Balance Equation**

*Thermal energy balance in terms of internal energy U per unit mass of reaction mixture .

$$\rho \frac{D\widehat{U}}{Dt} = -\underline{\nabla}.q - p\underline{\nabla}.\underline{v} + \underline{p}:\underline{\nabla v} \quad (191)$$

$$\underline{p}:\underline{\nabla v} = \sum_{i} \sum_{j} P_{ij} \frac{\partial v_i}{\partial x_j}$$

The enthalpy can be written as H=U +PV

$$\frac{D\widehat{H}}{Dt} = \frac{D\widehat{U}}{Dt} + P\frac{D\widehat{V}}{Dt} + \widehat{V}\frac{DP}{Dt}$$

$$\rho \frac{D\widehat{H}}{Dt} = \rho \frac{D\widehat{U}}{Dt} + p\underline{\nabla}.\underline{v} + \frac{DP}{Dt} = -\underline{\nabla}.q + \underline{p}:\underline{\nabla v} + \frac{DP}{Dt} \quad (from\ 191)$$

$$\rho\widehat{H} = \varepsilon \sum_{j=1} C_j \widehat{H}_j + (1-\varepsilon)H_s \quad (193)$$

Differentiate (193) w.r.t. time

$$\rho \frac{\partial \widehat{H}}{\partial t} = \epsilon \sum_{j=1}^{s} \frac{\partial C_j}{\partial t} \overline{H_j} + \epsilon \sum_{j=1}^{s} C_j \frac{\partial \overline{H}_j}{\partial t} + (1-\varepsilon)\frac{\partial H_s}{\partial t}$$

Note:

$$\begin{aligned}
\overline{H}_j\left(T, P, C_j\right) &= \epsilon \sum_{j=1}^{s} \frac{\partial C_j}{\partial t} \overline{H}_j \\
&\quad + \varepsilon \sum_{j=1}^{s} C_j \left[\left(\frac{\partial \overline{H}_j}{\partial T}\right)_{P,C} \frac{\partial T}{\partial t} + \left(\frac{\partial \overline{H}_j}{\partial P}\right)_{T,C} \frac{\partial P}{\partial t} + \sum_{k=1}^{s} \left(\frac{\partial \overline{H}_j}{\partial C_k}\right)_{P,T} \frac{\partial C_k}{\partial t}\right] \\
&\quad + (1-\varepsilon)\frac{\partial H_s}{\partial T}\frac{\partial T}{\partial t}
\end{aligned}$$

$$\begin{aligned}
\rightarrow \rho \frac{\partial \widehat{H}}{\partial t} &= \epsilon \sum_{j=1}^{s} \frac{\partial C_j}{\partial t} \overline{H}_j + \left[\sum_{j=1}^{s} \epsilon C_j \left(\frac{\partial \overline{H}_j}{\partial T}\right)_{P,C} + (1-\varepsilon)\frac{\partial H_s}{\partial T}\right]\frac{\partial T}{\partial t} \\
&= \epsilon \sum_{j=1}^{s} \frac{\partial C_j}{\partial t} \overline{H}_j + \left[\sum_{j=1}^{s} \epsilon C_j C_{pj} + (1-\varepsilon)C_{ps}\right]\frac{\partial T}{\partial t} \quad (194)
\end{aligned}$$

$$\rightarrow \rho \frac{\partial \hat{H}}{\partial t} = \sum_{j=1}^{s}\left[-\underline{\nabla}.\underline{J}_{j,e} + \sum_{i=1}^{m}\rho_b S_{gi}\alpha_{ij}\hat{r}_i\right]H_j + \left[\sum_{j=1}^{s}\varepsilon C_j C_{pj} + (1-\epsilon)C_{ps}\right]\frac{\partial T}{\partial t}$$

$$= -\sum_{j=1}^{s}\left(\underline{\nabla}.\underline{J}_{j,e}\right)\bar{H}_j + \sum_{i=1}^{m}\rho_b S_{gi}\,\hat{r}_i\sum_{j=1}^{s}\alpha_{ij}\,H_j$$

$$+ \left[\sum_{j=1}^{s}\varepsilon C_j C_{pj} + (1-\epsilon)C_{ps}\right]\frac{\partial T}{\partial t} \quad (195)$$

In above, we recognize that $\sum_{j=1}^{s}\alpha_{ij}\,H_j = \Delta H_j$ is the heat of ith reaction

$$C_p\frac{\partial T}{\partial t} = \underline{\nabla}.\left(\sum_{j=1}^{s}\underline{J}_{j,e}\,\bar{H}_j\right) + \sum_{i=1}^{m}\rho_b S_{gi}\,\hat{r}_i\Delta H_j - \underline{\nabla}.\underline{q} + \frac{\partial p}{\partial t} \quad (196)$$

The energy flux in multicomponent mixture can be written as

$$\underline{q} = -Ke\underline{\nabla}T + \sum_{j=1}^{s}\underline{J}_{j,e}\,\bar{H}_j \qquad (197)$$

Substitute (197) in (196, neglecting contribution from $\partial p/\partial t$ we get

$$C_p\frac{\partial T}{\partial t} = \underline{\nabla}.Ke\underline{\nabla}T + \sum_{i=1}^{m}\rho_b S_{gi}\,\hat{r}_i\Delta H_j \quad (198)$$

Equation (190) and (198) are the pair of PDEs that represent concentration and temperature distribution in a porous catalytic system.

The initial conditions are

t=0, $\underline{C}(\underline{r},0) = \underline{C_0}(\underline{r})$, $r \in \Omega$; $T(\underline{r},0)=T_0(\underline{r})$ $r \in \Omega$ (199)

The boundary conditions are

$$\epsilon K_{c,j}\left(C_{j,f} - C_{j,s}\right) = \sum_{k=1}^{s} D_{jk,e}\underline{n}\underline{\nabla}\,C_k|_{\partial\Omega} \quad j = 1,2 \ldots s$$

$$h\left(T_f - T_s\right) = K_e\underline{n}\underline{\nabla}T|_{\partial\Omega}$$

Note: At steady state condition, the eqn (190) and (198) may yield multiple steady states!

*****Non-catalytic solid-fluid reaction**

*Here we are concerned with reaction between solid and fluids in which solid participates as reactant.

A (gas) +bB (solid) → cC (gas) + dD (solid)

There are several examples of these reactions such as coal gasification, iron production, ore processing.

Ex: Iron production: $Fe_2O_{3(s)} + 3H_{2(g)} = 2Fe + 3H_2O$

Nickel production: $NiO + CO = Ni + CO_2$

Coal gasification: $C + ½\,O_2 = CO$

$$\frac{\partial}{\partial t}(\epsilon C_j) + \underline{\nabla}.\underline{N}_j = \rho_b \sum_{i=1}^{m} S_g \alpha_{ij} \hat{r}_i\,(C,T) \quad (201)$$

ρ_b, t, S_g are function of time

The continuity eqn for the solid component

$$\frac{\partial C_s}{\partial t} = \rho_b \sum_{i=1}^{m} S_g \alpha_{ij} \hat{r}_i\,(C,T) \quad (202)$$

C_s = concentration of solid per unit particle volume.

The energy balance eqn may be written as

$$C_p \frac{\partial T}{\partial t} = \underline{\nabla}.Ke\underline{\nabla}T + \rho_b \sum_{i=1}^{m} (-\Delta H_j) S_g \alpha_{ij} \hat{r}_i\,(C,T) \quad (203)$$

Cp represents average volumetric specific heat of the particle. Since rate of reaction inside particle is higher than rate of diffusion, it is reasonable to assume pseudo-steady state.

$$\frac{\partial C_i}{\partial t} = 0\ in\ 201$$

$$C_j \frac{\partial \varepsilon}{\partial t} + \underline{\nabla}.\underline{N}_j = \rho_b \sum_{i=1}^{m} S_{gi} \alpha_{ij} \hat{r}_i\,(C,T) \quad (204)$$

Porosity $\epsilon = f(s)$

$$C_j f'(C_s) \frac{dC_s}{dt} + \underline{\nabla}.\underline{N}_j = \rho_b \sum_{i=1}^{m} S_{gi} \alpha_{ij} \hat{r}_i\,(C,T)$$

From (202)

$$C_j f'(C_s) \rho_b \sum_{i=1}^{m} S_{gi} \alpha_{ij} \hat{r}_i\,(C,T) + \underline{\nabla}.\underline{N}_j = \rho_b \sum_{i=1}^{m} S_{gi} \alpha_{ij} \hat{r}_i\,(C,T)$$

$$\underline{\nabla}.\underline{N}_j = \rho_b \sum_{i=1}^{m} \left[\alpha_{ij} - C_j f'(C_s)\right] S_{gi} \alpha_{ij} \hat{r}_i(C,T) \quad (205)$$

If there is an ash layer (non-reactive solid), eqn 201 becomes

$$\epsilon \frac{\partial C_j}{\partial t} + \underline{\nabla}.\underline{N}_j = 0 \quad (206)$$

Let’s consider a special case below

***the Shrinking core model**

*A spherical particle of radius R_0 is participating in a single solid-fluid reaction [αS+A$\rightarrow$P]

*The particle has initial radius R_0 and is assumed to have retreated to radius R(t) during the course of reaction.

*the continuity equation for the ash layer under PSSH becomes

$$\underline{\nabla}.\underline{N} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 De\frac{\partial C_A}{\partial r}\right) = 0 \quad (207)$$

With B.Cs

$$r = R_0, \qquad De\frac{\partial C_A}{\partial r} = k_c\left(C_{Af} - C_A\right)$$

$$r = R(t), \qquad De\frac{\partial C_A}{\partial r} = \hat{k}C_A C_{S_o} \; (208)$$

To calculate R(t), we consider the surface of particle as : $g(\underline{x},t)=x_i x_i - R^2(t)$

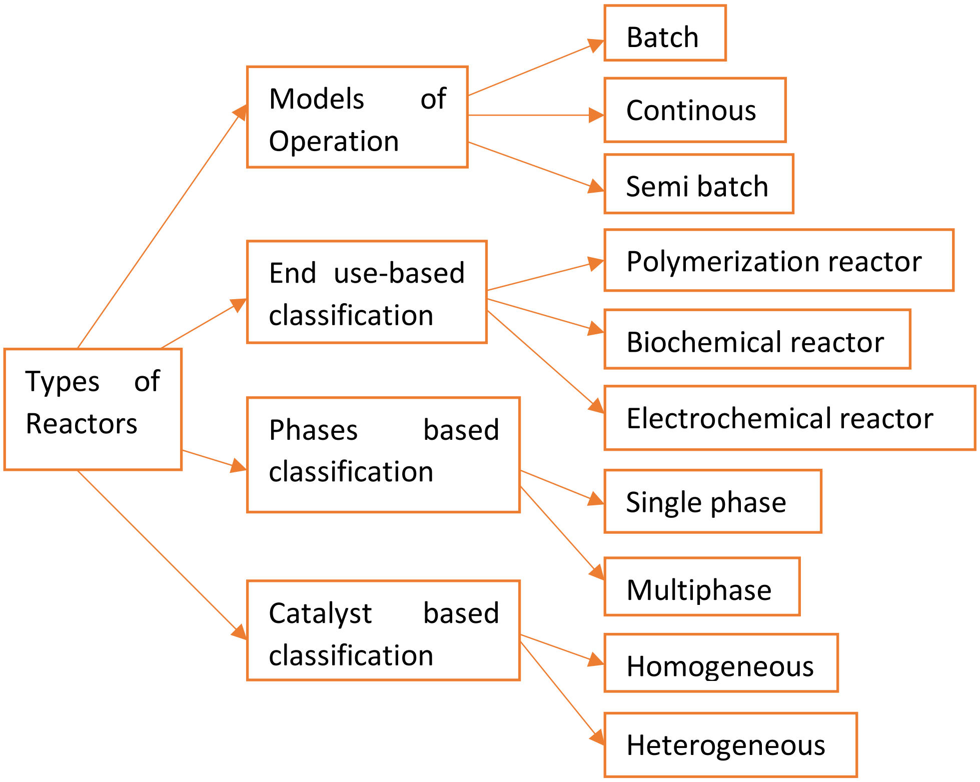
$$\frac{\partial g}{\partial t} = -2R\frac{dR}{dt} = \alpha\hat{k}C_A C_{S_o}\,|\nabla g| = \alpha\hat{k}C_A C_{S_o}\,2R$$

$$\frac{dR}{dt} = -\alpha\hat{k}C_A C_{S_o} \;\; (209)$$

Solve (207) and (208) to obtain C(r,R(t)), then substitute in (209) to solve for R(t)

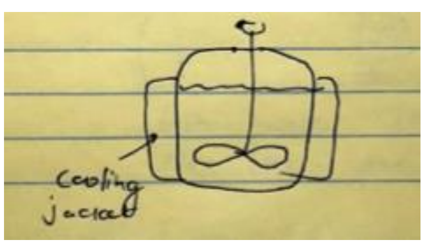
*****Chemical Reactor Analysis**

*There are various types of chemical reactors in the chemical process industries



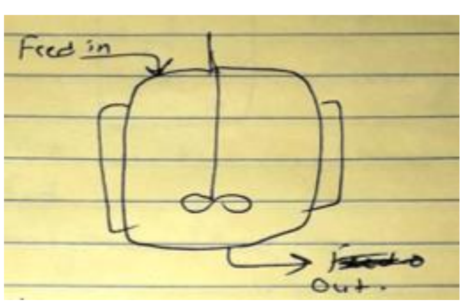
Ex:

1. Batch Reactor

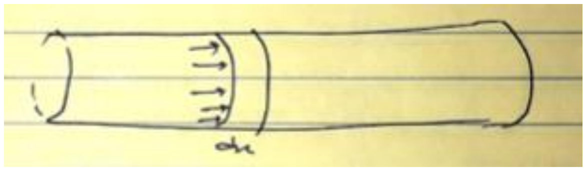


2. Continuous Reactor

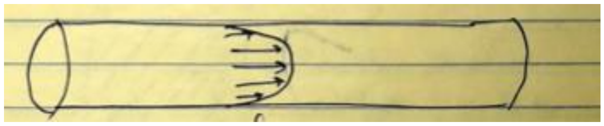
Continuous stirred tank reactor



Plug Flow Reactor : every fluid element in the cross-section of tube flow at the same velocity.

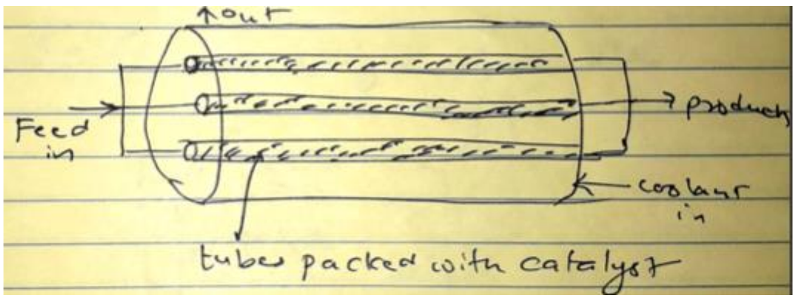


Tubular Reactor: Laminar flow reactor. The flow profile is laminar.

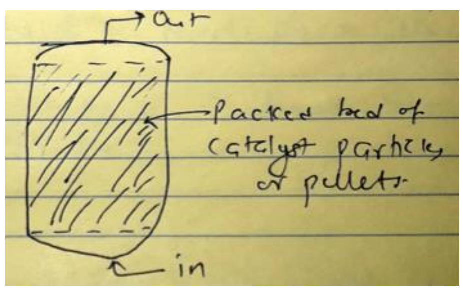


4. Catalytic Reactor

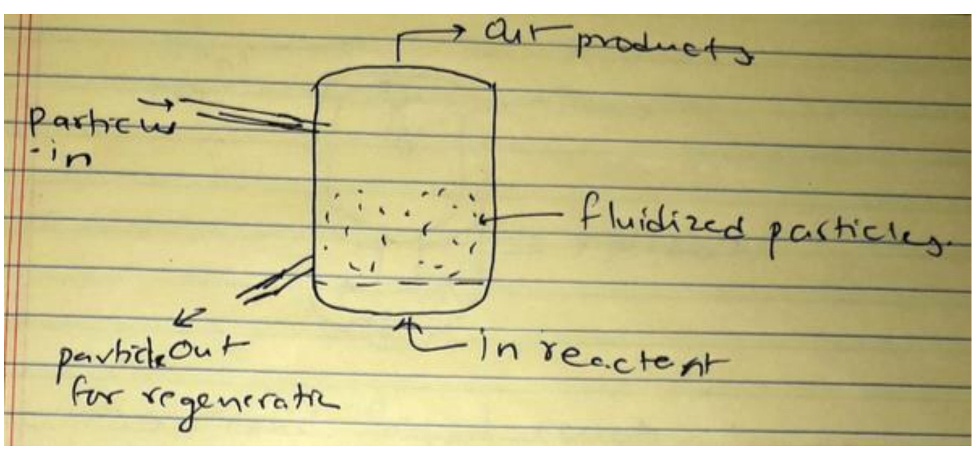
Tubular Reactor



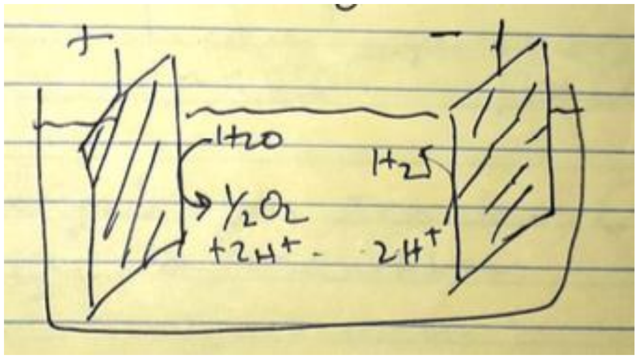
Packed bed or fixed bed reactor



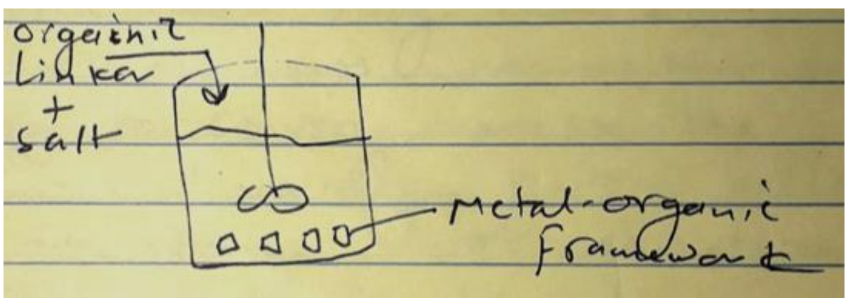
Fluidized Bed Reactor



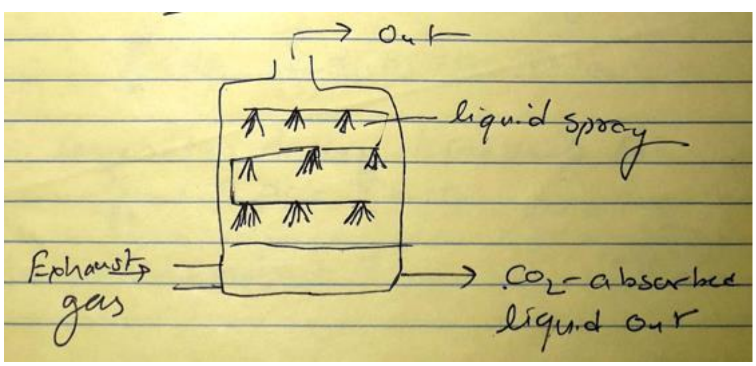
Electrocatalytic Reactions



5. Non-catalytic multiphase reactors
   *gas-solid or liquid-solid reactor
   *Solid-liquid reactor
   Ex: reactive crystallization



*Liquid-gas reactor (spray reactor). Ex: CO_2 scrubber



*Equations for the analysis of reactors can vary from macroscopic balances of mass, momentum and energy to solving microscopic balances involving partial differential equation for variation in catalyst or reaction layer.

*Consider m reactions among s species

$$\sum_{j=1}^{s} \alpha_{ij} A_j = 0, i = 1,2 \dots m$$

*Concentration of mass/continuity eqn

$$\frac{\partial C_j}{\partial t} + \underline{\nabla}.\underline{N}_j = \sum_{i=1}^{m} \alpha_{ij} r_j = 0, j = 1,2 \dots s \quad (210) - constant\ V$$

For heterogeneous reactions: $r_j = \rho_b S_{gi} \hat{r}_i$

The total molar flux of A_j in a multicomponent mixture is given by the Stetan-Maxwell equations

$$-C\underline{\nabla}x_j = \sum_{k=1}^{s} \frac{x_k \underline{N}_j - x_j N_k}{D_{jk}} \quad (211)$$

D_{jk} is the binary diffusion coefficient of A_j w.r.t A_k

Define an effective binary diffusion coefficient

$$\frac{1}{D_{jm}} = \sum_{k=1}^{s} \frac{x_k}{D_{jk}} \quad (212)$$

Substitute (212) in (211)

$$\underline{N}_j = x_j \sum_{k=1}^{s} \frac{D_{jm} \underline{N}_K}{D_{jk}} - C\, D_{jm} \underline{\nabla} x_j = x_j \underline{N} - C D_{jm} \nabla x_j$$

*****Macroscopic Balance of Mass in Reactors**

*****In so far, we considered microscopic mole balance at the site of reaction either on a catalyst surface or in a small homogeneous volume of a reactor.

*Often it is required to formulate a macroscopic balance of mass in a reactor

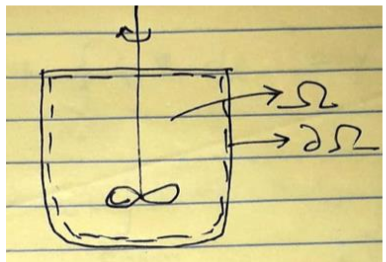
*Assume the reactor domain is Ω with $\partial\Omega$ as its boundary surface which contains $\partial\Omega\text{in}$ surface to let material into the reactor and $\partial\Omega\text{out}$ through which reacted material leaves.

*Integrate (187) for porous domain or (210) for homogeneous domain

$$\frac{d}{dt}\int_{\Omega} C_j dV + \oint_{\partial\Omega} C_j V.dA = \int_{\Omega} \sum_{i=1}^{m} \alpha_{ij} r_i dV \quad (213)$$

Case 1: Macroscopic mole balance for batch reactor

$$\frac{d}{dt}\int_{\Omega} C_j dV + \int_{\Omega} \underline{\nabla}.\underline{N}_j dV = \int_{\Omega} \sum_{i=1}^{m} \alpha_{ij} r_i dV$$



In a well-stirred reactor C_j is uniform such that $C_j dV \sim dC_j V$

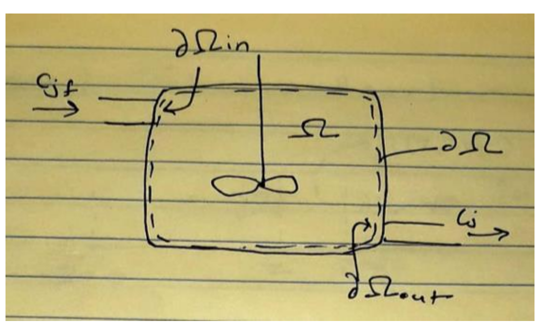
$$\frac{d}{dt}(VC_j) = V\sum_{i=1}^{m} \alpha_{ij} r_i \quad (214)$$

The volume change w.r.t t is given by

$$\frac{dlnV}{dt} = \sum_{i=1}^{m}\sum_{j=1}^{s} \alpha_{ij} \overline{V}_j r_i$$

Case 2: Macroscopic mole balance for continuous stirred tank reactor (CSTR)

$$\frac{d}{dt}\int_{\Omega} C_j dV + \int_{\Omega} \underline{\nabla}.\underline{N}_j dV = \int_{\Omega} \sum_{i=1}^{m} \alpha_{ij} r_i dV$$



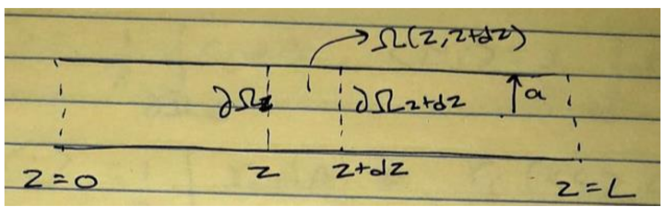
$$\frac{d}{dt}(VC_j) + \oint_{\partial\Omega} C_j V.\,dA = V\sum_{i=1}^{m} \alpha_{ij} r_i$$

$$\frac{d}{dt}(VC_j) + C_{j,f}\oint_{\partial\Omega\mathrm{in}} \underline{V}.\,dA + C_j\oint_{\partial\Omega\mathrm{out}} \underline{V}.\,dA = V\sum_{i=1}^{m} \alpha_{ij} r_i$$

$$\frac{d}{dt}(VC_j) - q_{in}C_{j,f} + q_{out}C_j = V\sum_{i=1}^{m} \alpha_{ij} r_i \quad (215)$$

The quantity V/q $=\tau$ is known as the nominal holding time or the residence time of the reactor. In biological reactor, D=1/ τ is referred to as the dilution rate.

Case 3: Tubular Reactor



*For tubular reactor, it is common to integrate (210) over Ω which is the region boundary by the reactor all and the cross-sections at z and z+dz

*Reaction mixture flows in +z direction so that $\partial\Omega\mathrm{in} = \partial\Omega\mathrm{z}$ and $\partial\Omega\mathrm{out} = \partial\Omega(\mathrm{z} + \mathrm{dz})$

*Macroscopic balance over $\Omega\ (\mathrm{z}, \mathrm{z} + \mathrm{dz})$ gives

$$\frac{\partial}{\partial t}\int_{\Omega} C_j dV + \int_{\partial\Omega\mathrm{z}} \underline{N}_j dV + \int_{\partial\Omega(\mathrm{z+dz})} \underline{N}_j dV = \int_{\Omega(\mathrm{z,z+dz})} \sum_{i=1}^{m} \alpha_{ij} r_i dV$$

$$\frac{\partial}{\partial t} dz \int_{\Omega} C_j dA + \int_{\partial\Omega\mathrm{z}} \underline{N}_j dV + \int_{\partial\Omega(\mathrm{z+dz})} \underline{N}_j dV = dz \int_{\Omega(\mathrm{z,z+dz})} \sum_{i=1}^{m} \alpha_{ij} r_i dA \quad (216)$$

Define, area-averaged quantities

$$< C_j > = \frac{1}{A}\int_{\partial\Omega\mathrm{z}} C_j dA$$

$$< N_j > = \frac{1}{A} \int_{\partial\Omega z} N_j dA$$

$$< r_j > = \frac{1}{A} \int_{\partial\Omega z} r_i dA \ \sim \ r_i\,(< C >, < T >)$$

Divide (216) by dz and take lim (dz→0)

$$\frac{\partial < C_j >}{\partial t} + \frac{\partial}{\partial z} < N_j > = \sum_{i=1}^{m} \alpha_{ij} r_i(< C >, < T >) \ \ (217)$$

The area-averaged flux can be expressed as

$$< N_j > = < v_z C_j - D_{jm} \frac{\partial C_j}{\partial z} > \ \ (218)$$

G.I Taylor (1953) considered dispersion of dye tracer in a fully developed flow

$$v_z = v_{zmax}\left(1 - \frac{r^2}{a^2}\right)$$

“telephone wire singing in the wind” Taylor analysis resulted in

$$< N_{j,z} > \sim \ < v_z >< C_j > - D_{jm} \frac{\partial < C_j >}{\partial z} \ \ (219)$$

$$D = D_{jm} + \lambda \frac{a^2 {v_{zmax}}^2}{192 D_{jm}} \ \ \ (220)$$

D is the dispersion coefficient

λ is the shape factor of tube

$<v_z> = v_{zmax}/2$

Substitute (219) in (217)

$$\frac{\partial < C_j >}{\partial t} + < v_z > \frac{\partial}{\partial z} < C_j > = D \frac{\partial^2 < C_j >}{\partial z^2} + \sum_{i=1}^{m} \alpha_{ij} r_i(< C >, < T >) \ \ (221)$$

This is axial dispersion model or macroscopic mole balance for tubular reactor

Case 3-1: Plug Flow reactor

Plug Flow model is obtained when $\frac{\partial^2 < C_j >}{\partial z^2} = 0$ in (221)

Initial condition $< C_j(z, 0) > = \ C_{jo}(z)$

Boundary conditions for the tubular reactor require some discussion. They are popularly known as Danckwerts boundary conditions.

At z=0

$$D_j \frac{d < C_j >}{dz} - < v_z >< C_j > = \ - < v_z > C_f \quad (222)$$

No diffusive flux or dispersion before the entrance of the tube

At z=L, $\frac{d < C_j >}{dz} = 0 \quad (223)$

This is the most debated B.C in chemical engineering.